# Predictive Modeling of Cattle Metabolic Disease Risk Using Bayesian Logistic Regression


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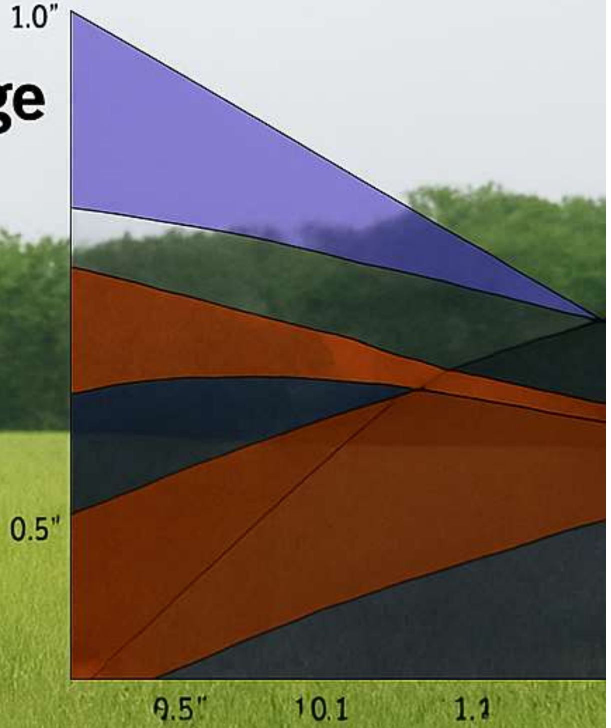
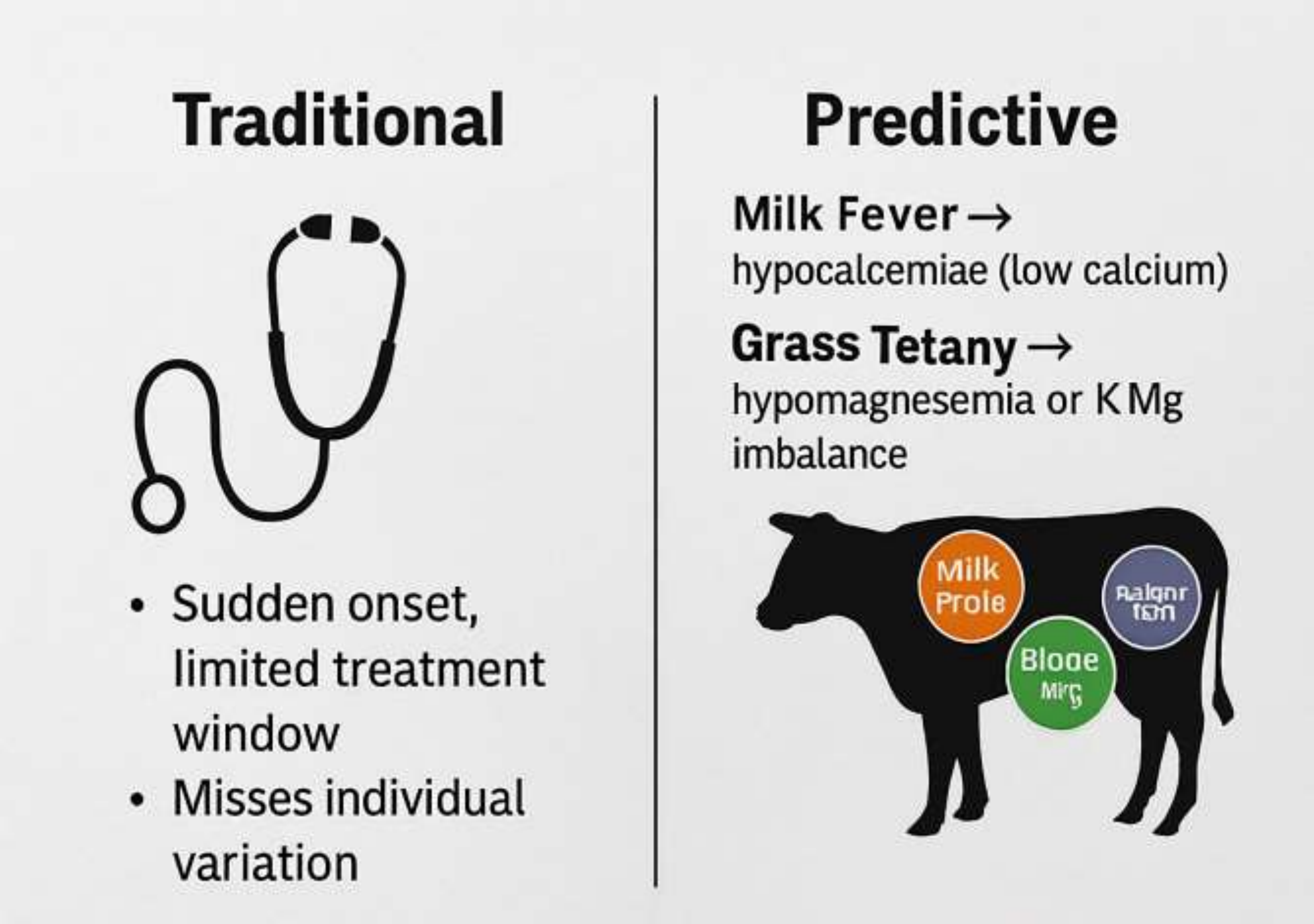
**Data Science Thesis | Warren Wilson College**

# Research Problems

- These diseases strike suddenly, and traditional diagnostics rely on thresholds or visual symptoms.
- My research addresses the gap by modeling individualized risk probabilities
- Just as the derivative reveals instantaneous change, our Bayesian Logistic Regression models aim to detect early shifts in health trajectories.



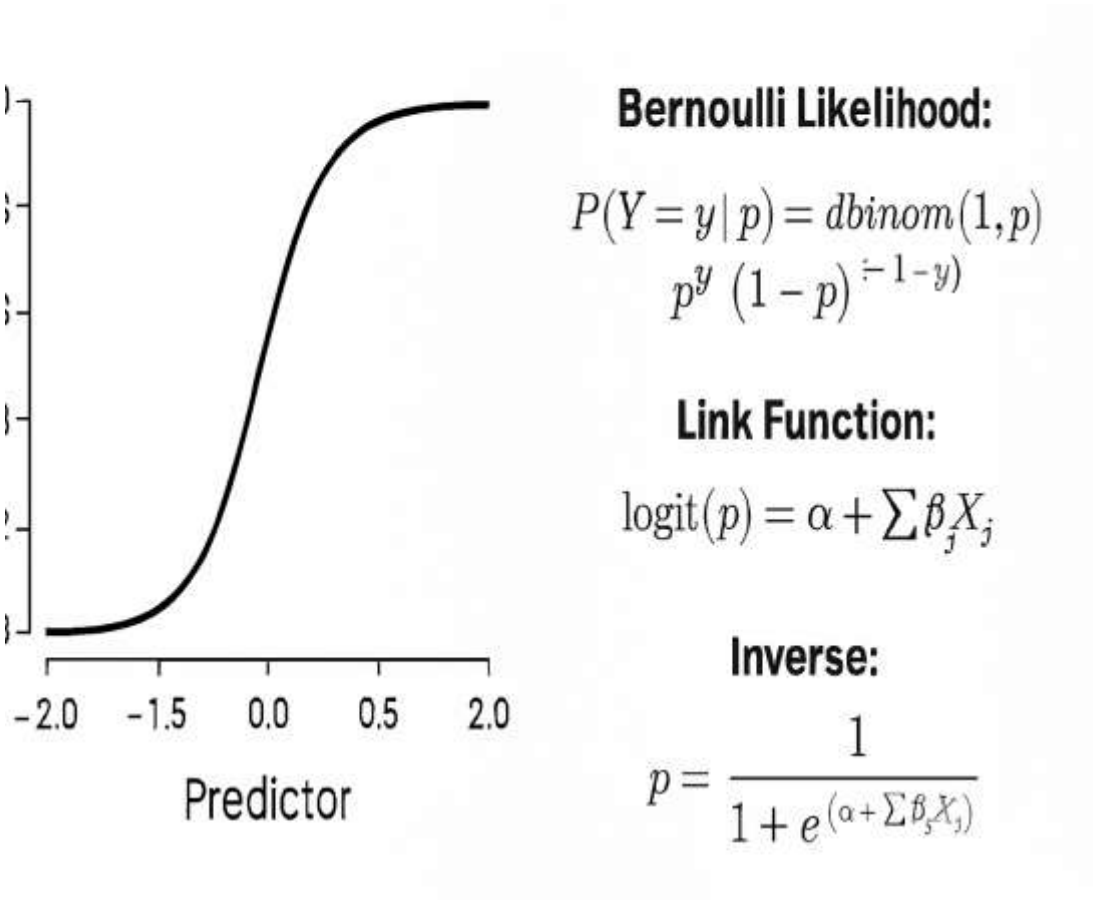
# Objective & Data Sources

- **Objective:**
  To develop interpretable machine learning models that detect early metabolic vulnerability by integrating mineral thresholds, behavioral indicators, and temporal dynamics.
- **Data Source:**
  We use longitudinal cattle health data from Warren Wilson College's working farm, including:
- **Biochemical assays** (ICP-OES, AAS) for calcium, magnesium, and potassium levels
- **Behavioral metrics** such as chute score and exit velocity
- **Real-world conditions** reflecting ecological variability and physiological stress

These multimodal inputs allow us to model biological systems as dynamic structures — where gradients, resistance, and inflection points reveal early signs of dysfunction.

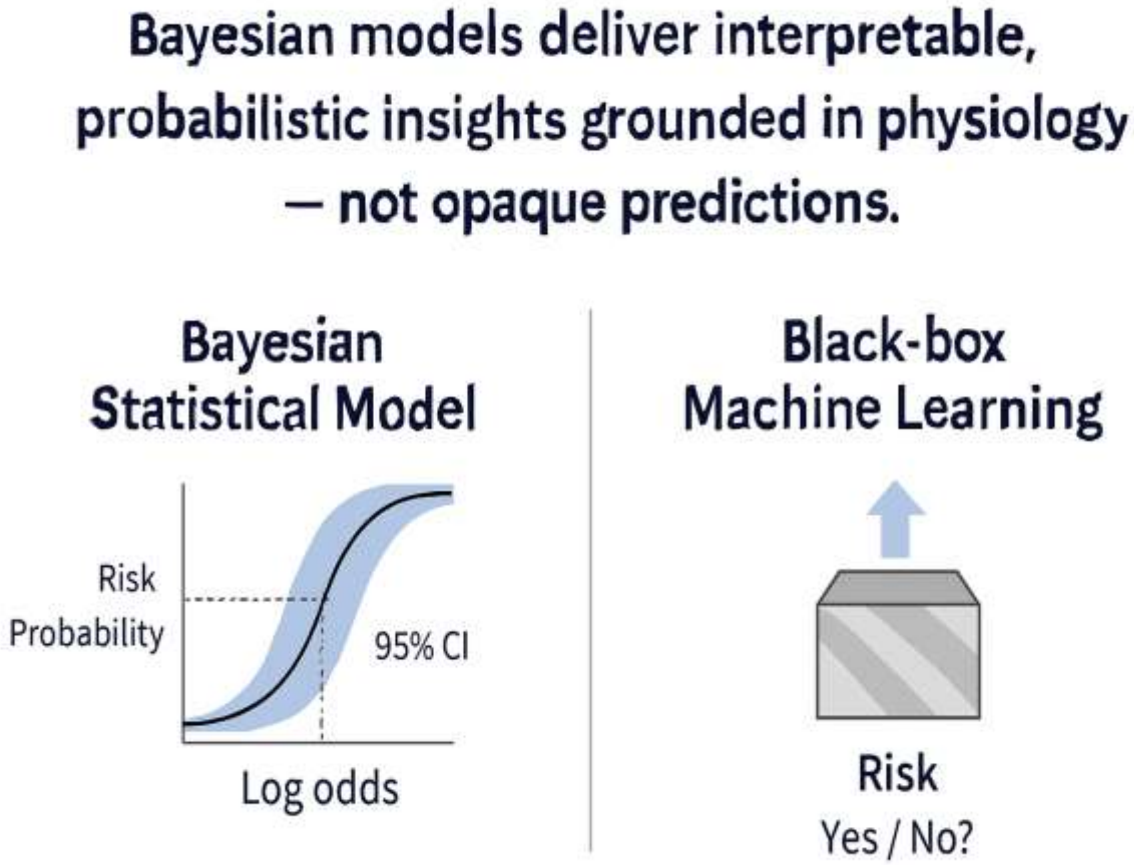
# Methodology Overview

- **Model:** Bayesian logistic regression (rethinking::map)
- **Likelihood:** Bernoulli - $L(p) = dbinom(1, p) = p^1 \left(1 - p\right)^0$ defines outcome probabilities
- **Link Function:** Logistic - $\mathrm{logit}(p) = \alpha + \sum_j \beta_j X_j$ transforms predictors into interpretable risk estimates
- **Predictors:** Standardized calcium, magnesium, potassium, chute score, exit velocity, encoded hair whorl
- **Labels:** Synthetic, physiologically grounded rules: • Mg < 1.6 • K:Mg > 3.0 • Ca < 8.0
- **Validation:** 4-fold cross-validation (folds: 10, 10, 10, 9) with fold-specific GT and MF models
- **Modeling Approach:** We apply Bayesian logistic regression with physiologically grounded labels, integrating mineral and behavioral predictors through cross-validated models



# Why Bayesian (and not black-box ML?)

- **Probabilities, not flags:**
  Outputs are continuous risk probabilities per cow, not binary yes/no labels.
- **Uncertainty is explicit** :
  Posterior distributions and HPDIs quantify confidence, not just point estimates.
- **Interpretability is physiological:**
  Coefficients map directly to mineral thresholds and behavioral indicators.
- **Data reality is respected:**
  Bayesian methods handle small, imbalanced datasets without sacrificing rigor.
- **From Physiology to Probabilities:**
  Bayesian models deliver interpretable, probabilistic insights grounded in physiology — not opaque predictions."



# Results Overview

## ROC Curve Performance

- **ROC:**
  - *Milk Fever (MF)* curve shows stronger discrimination — clear separation from random classifier
  - *Grass Tetany (GT)* curve weaker — reflects challenge of rare-event prediction

## Accuracy:

- **Model-based predictions:**
  - GT accuracy = **84.6%**
  - MF accuracy = **89.7%**

- **ROC Curve Insights:**
  Overall, ROC curves highlight the model's ability to discriminate risk, with Milk Fever showing strong separation and Grass Tetany reflecting the inherent challenge of rare-event prediction

- **Rule-based flags (thresholds):**
  - GT accuracy = **97.4%**
  - MF accuracy = **100%**
- *Insight:* Deterministic flags are highly accurate but lack nuance. Model predictions add uncertainty quantification and better generalization.

## Coefficient Effects (Posterior HDPI)

- **Lower Ca and Mg** → strong increase in predicted risk
- **Chute and Exit scores** → weak associations with stress physiology
- **Hair whorl position** → small but directional behavioral effect

# Confusion Matrix: Diagnostic Performance for GT and MF

This table compares the cows flagged as ‘at risk’ by our model against their actual health outcomes. It shows how well the risk-flagging aligns with reality.”

## Grass Tetany (GT)

- **Flagged Positive (5 cows):** All 5 were truly positive — no false alarms.
- **Not Flagged Positive (1 cow):** One sick cow was missed.
- **Not Flagged Negative (33 cows):** 33 healthy cows correctly identified.
- **Summary:** GT flagging is **highly specific** (no false positives) but missed one true case, showing strong but not perfect sensitivity.
- **GT:** High specificity, strong sensitivity, missed 1 sick cow.

## Milk Fever (MF)

- **Flagged Positive (35 cows):** All 35 were truly positive — no false alarms.
- **Not Flagged Positive (0 cows):** No sick cows were missed.
- **Not Flagged Negative (4 cows):** Four healthy cows were incorrectly flagged as sick.
- **Summary:** MF flagging is **perfectly sensitive** (caught all sick cows) but less specific, since some healthy cows were misclassified.
- **MF:** Perfect sensitivity, lower specificity, 4 false positives.

# Diagnostic Metrics for Model Evaluation

**Sensitivity (True Positive Rate)**
**Formula:**

$$\text{Sensitivity} = \frac{\mathit{True\ Positives\ (TP)}}{\mathit{True\ Positives\ (TP)} + \mathit{False\ Negatives\ (FN)}}$$

**Interpretation:**
Sensitivity measures how well the model identifies sick cows. It answers:

*"Out of all the cows that are actually sick, how many did the model correctly flag?"* A sensitivity of 1 means the model flagged every sick cow correctly.

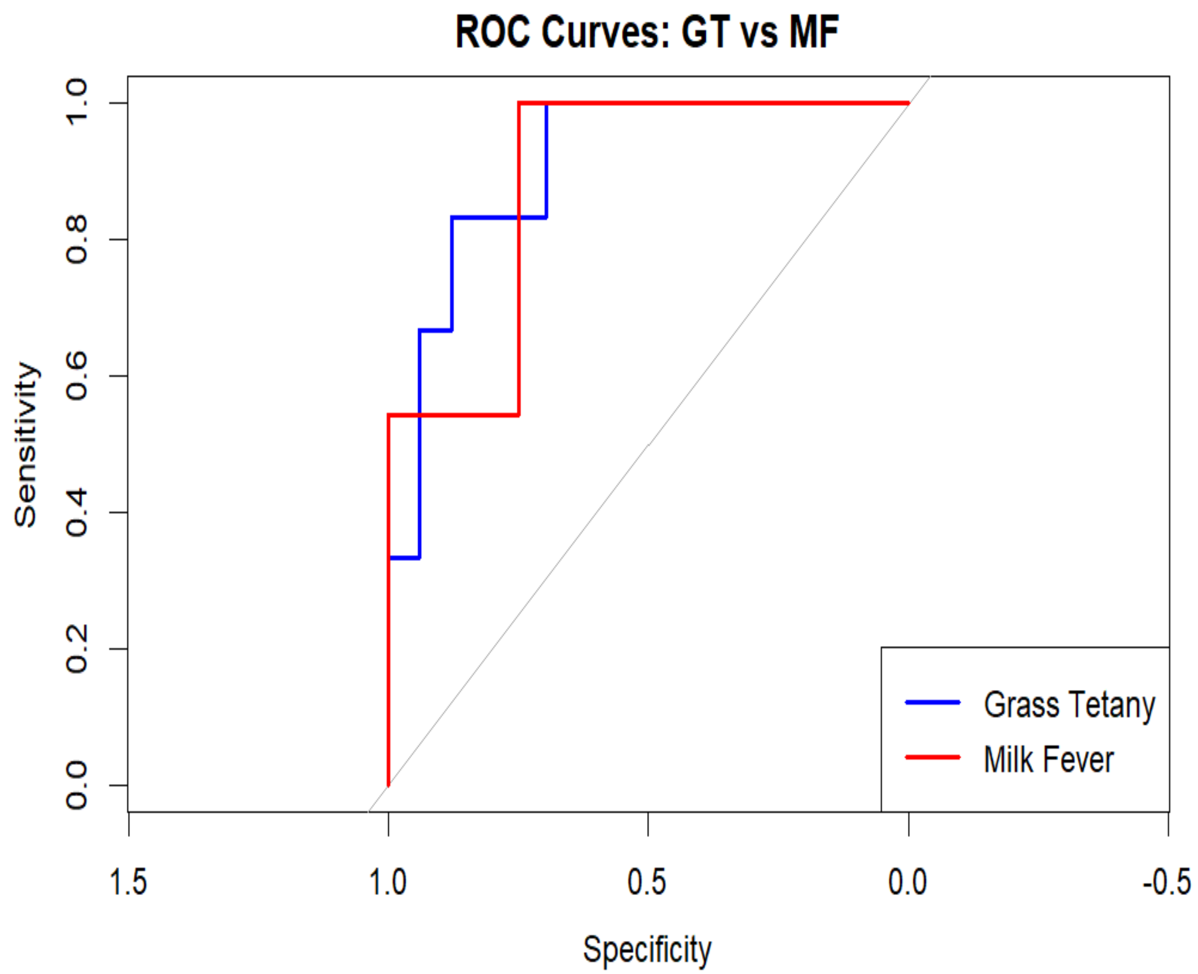
**Specificity (True Negative Rate)**
**Formula:**

$$\text{Specificity} = \frac{\mathit{True\ Negatives\ (TN)}}{\mathit{True\ Negatives\ (TN)} + \mathit{False\ Positives\ (FP)}}$$

**Interpretation:**
Specificity measures how well the model identifies healthy cows. It answers:

*"Out of all the cows that are actually healthy, how many did the model correctly leave unflagged?"* A specificity of 1 means the model avoided false alarms and correctly identified all healthy cows.

# ROC Curves – Sensitivity vs Specificity



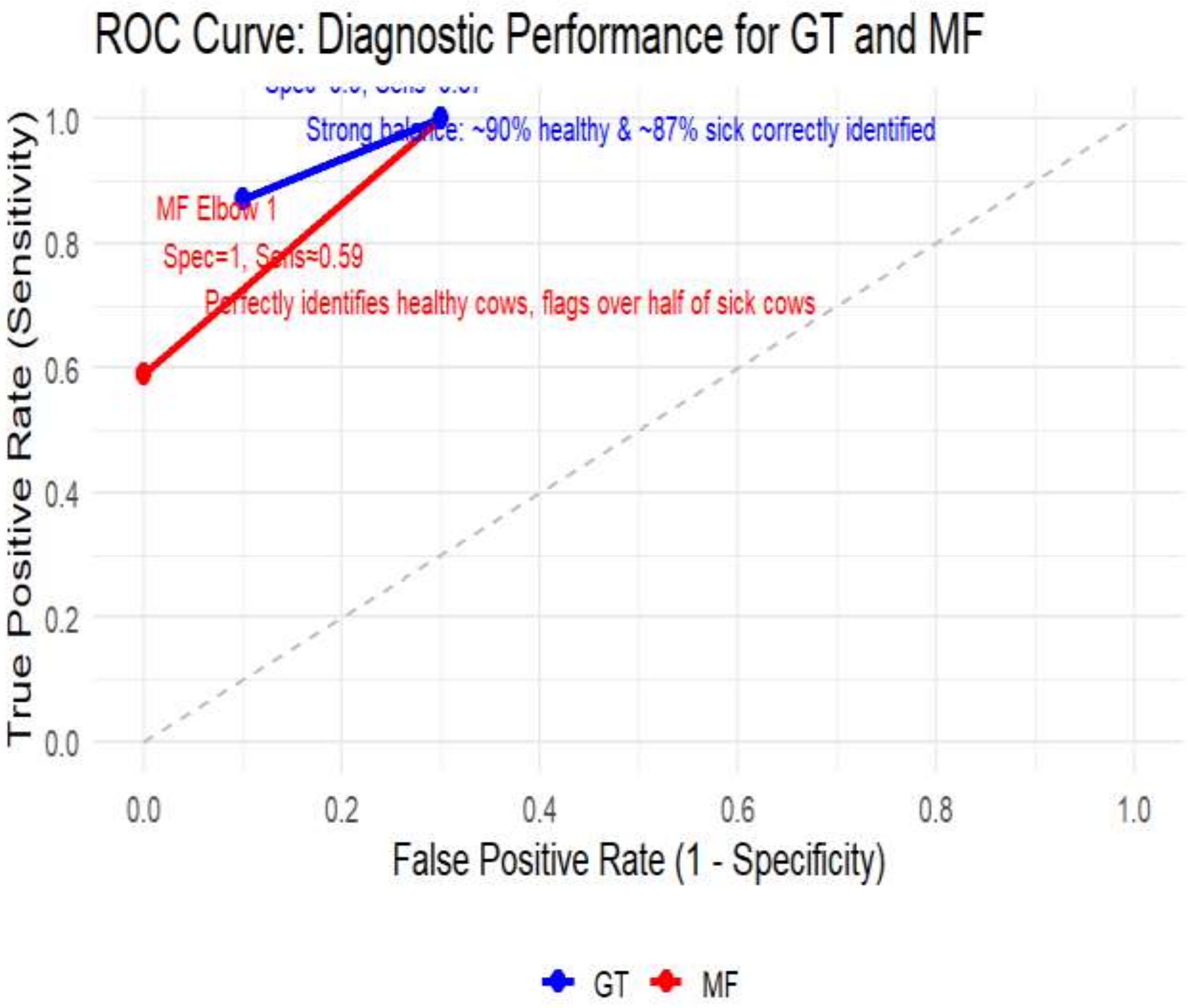

- **Sensitivity (True Positive Rate)**: Measures how well the model identifies sick cows. A sensitivity of 1 means the model flags every sick cow correctly.

- **Specificity (True Negative Rate)**: Measures how well the model identifies healthy cows. A specificity of 1 means the model avoids false alarms and correctly identifies all healthy cows.

- **False Positive Rate (1 - Specificity)**: The proportion of healthy cows incorrectly flagged as sick.

- **Elbow Points**: Represent optimal trade-offs between sensitivity and specificity — where the model performs best diagnostically.

# ROC Curve: Diagnostic Accuracy of Milk Fever vs Grass Tetany




**MF Elbow 1 (Red, Spec = 1, Sens ≈ 0.59):** Perfectly identifies healthy cows, but misses ~40% of sick cows.

**MF Elbow 2 (Red, Spec ≈ 0.7, Sens = 1):** Flags all sick cows, correctly identifies 70% of healthy cows.

**GT Elbow 1 (Blue, Spec ≈ 0.9, Sens ≈ 0.87):** Strong balance ~90% healthy and ~87% sick cows correctly identified.

**GT Elbow 2 (Blue, Spec ≈ 0.7, Sens = 1):** Flags all sick cows, correctly identifies 70% of healthy cows.

**Random Classifier (Dashed line):** Baseline performance, showing chance-level separation.

So overall, the red MF curve emphasizes trade-offs between strict specificity and full sensitivity, while the blue GT curve demonstrates a more balanced diagnostic profile. The dashed line shows what random guessing would look like, and both tests clearly outperform it

# Conclusion — Key Narrative

- **Bayesian predictive modeling enhances herd health diagnostics** by combining accuracy, interpretability, and uncertainty quantification.
- **Milk Fever:** Strong, reliable predictions with clear physiological grounding.
- **Grass Tetany:** Rare-event challenges but still reveals validated biological signals.
- **Key takeaway:** Accuracy alone is insufficient — interpretability and equity are essential pillars.
- **Physiological insights:** Low calcium and magnesium are strong predictors; behavioral indicators add nuance.
- **Actionability:** Dashboard integration makes results usable for farmers and clinicians.
- **Broader impact:** Positions predictive modeling as a bridge between advanced analytics, real-world decision-making, and translational applications in human health.

Predictive modeling is not just a statistical tool — it’s a pathway to equity, physiology, and impact. – **David Iyileh**

# Citations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  *Discusses how machine learning and deep learning are applied in veterinary diagnostics, including biomarker-based disease prediction.*
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- *Lee, S. (2025). Precision–Recall Curve: A comprehensive guide. Number Analytics Blog. Retrieved from https://www.numberanalytics.com/blog/precision-recall-curve-comprehensive-guide →*
  *Explains PR curves and why they are critical for imbalanced datasets like cattle disease prediction.*

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